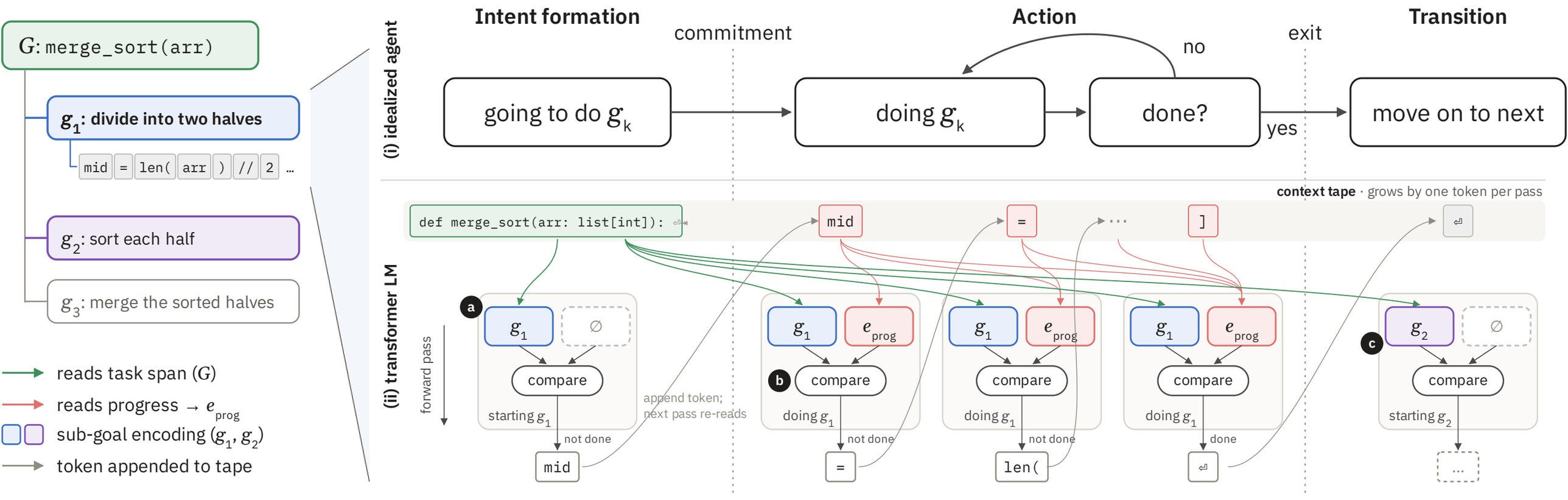

G: merge_sort(arr)
g1: divide into two halves
mid = len(arr) // 2 …
g2: sort each half
g3: merge the sorted halves
reads task span (G)
reads progress → e prog
sub-goal encoding (g1, g2)
token appended to tape
(i) idealized agent
Intent formation
commitment
Action
no
exit
Transition
going to do gk
doing gk
done?
yes
move on to next
(ii) transformer LM
context tape · grows by one token per pass
def merge_sort(arr: list[int]):
mid
=
…
]
forward pass
a
g1
∅
compare
starting g1
not done
mid
append token;
next pass re-reads
b
g1
eprog
compare
doing g1
not done
=
g1
eprog
compare
doing g1
not done
len(
g1
eprog
compare
doing g1
done
c
g2
∅
compare
starting g2
…